0. **Code:** bhagams4063 bhagam sai goutham test1.zip

1. test1/test1/Program.cs
2. test1/test2/Program.cs
3. test1/test3/Program.cs

1. **test1/test1/Program.cs**

```
namespace test1
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Enter number 1");
            int a = int.Parse(Console.ReadLine());
            Console.WriteLine("Enter number 2 ");
            int b = int.Parse(Console.ReadLine());
            Console.WriteLine("Enter number 3");
            int c = int.Parse(Console.ReadLine());

            int count = 0;
            double sum = 0;
            if (a ≠ 0 && b ≠ 0 && c ≠ 0)
            {
                int quotient1 = b / a;
                int remainder1 = b % a;
                if (quotient1 ≠ 0 && remainder1 ≠ 0)
                {
                    count++;
                    sum += remainder1;

                }
                int quotient2 = a / b;
                int remainder2 = a % b;
                if (quotient2 ≠ 0 && remainder2 ≠ 0)
                {
                    count++;
                    sum += remainder2;
```

A1

A2

this is not necessarily true: It's valid even when 1 of them is 0.

if(a != 0) {

A3

} if(b != 0) {

A3

1. **test1/test1/Program.cs**

```
            }
            int quotient3 = c / a;
            int remainder3 = c % a;
            if (quotient3 ≠ 0 && remainder3 ≠ 0)
            {
                count++;
                sum += remainder3;

            }
            int quotient4 = a / c;
            int remainder4 = a % c;
            if (quotient4 ≠ 0 && remainder4 ≠ 0)
            {
                count++;
                sum += remainder4;

            }
            int quotient5 = c / b;
            int remainder5 = c % b;
            if (quotient5 ≠ 0 && remainder5 ≠ 0)
            {
                count++;
                sum += remainder5;

            }
            int quotient6 = b / c;
            int remainder6 = b % c;
            if (quotient6 ≠ 0 && remainder6 ≠ 0)
            {
                count++;
```

1. **test1/test1/Program.cs**

```
                    sum += remainder6;

                }
            }
            if (count > 0)
            {
                double average = sum / count;
                Console.WriteLine(sum);
                Console.WriteLine(count);
                Console.WriteLine("The average is:" + average);
            }
            else
            {
                Console.WriteLine("The average cannot be calculated");
            }

        }
    }
}
```

A4

2. test1/test2/Program.cs

```
namespace test2
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int sum = 0;
            int count = 0;

            while (true)
            {
                Console.WriteLine("enetr a pair of integers");
                int a = int.Parse(Console.ReadLine());
                if (a == 0)
                {
                    break;
                }
                int b = int.Parse(Console.ReadLine());
                if (b == 0)
                {
                    break;
                }
                if (b ≠ 0 && a ≠ 0)
                {
                    int quotient1 = a / b;
                    int remainder1 = a % b;
                    if (remainder1 ≠ 0 && quotient1 ≠ 0)
                    {
                        sum += remainder1;
                        count++;
                    }
```

B1

B2

B3

2. **test1/test2/Program.cs**

```
                    int quotient2 = b / a;
                    int remainder2 = b % a;
                    if (remainder2 != 0 && quotient2 != 0)
                    {
                        sum += remainder2;
                        count++;
                    }

                }
                if (count > 0)
                {
                    double average = (double) sum / count;
                    Console.WriteLine("The average is" + average);

                }
                else
                {
                    Console.WriteLine("The average cannot be caluculated");
                }
            }
        }
    }
}
```

3. test1/test3/Program.cs

```
namespace test3
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int maxwithincreasing = int.MinValue;
            int minwithdecreasing = int.MaxValue;
            int count = 1; // the number of times the maximnum is set
            //int count2 = 0; // the number of times the mininmum is set
            int previous = 0;
            int cmax = 0; // same as count1
            int cmin = 0; // same as count2
            while (true)
            {
                Console.WriteLine("enetr the inputs:");
                int num = int.Parse(Console.ReadLine());
                if (num == previous)
                {
                    break;
                }

                if (previous ≠ 0)
                {
                    int lastdigitprev = previous % 10;
                    int lastdigitcurrent = num % 10;
                    if (lastdigitcurrent > lastdigitprev)
                    {
                        maxwithincreasing = Math.Max(maxwithincreasing,
                            previous);
```

C2

C3

C1

if(count>1)

C1

Use count!

prev can be 0.

need to consider current value.

Use if statement: if(maxwithincreasing < previous)? maxwith = previous

3. **test1/test3/Program.cs**

```
                count++;
                cmax = count;
                Console.WriteLine("max:" + count);

            }
else        if (lastdigitcurrent < lastdigitprev)
            {
                minwithdecreasing = Math.Min(minwithdecreasing,
                  previous);
                count++;
                cmin = count;
                Console.WriteLine("min:" + count);
            }

        }
        previous = num;

    }
    if (cmax ≠ 0 && cmin ≠ 0)
    {
        Console.WriteLine(cmax);
        Console.WriteLine(cmin);
        // the difference between the number of times max and min
          are set.
        // the number of values entered that lie between these min
          and max values.
        Console.WriteLine("count:" + (cmax - cmin));
    }
```

either/or

C3

use if statement.

need to consider current value.

C

count ++;

index/position entered.

C4

+1 ?

OK

absolute value |cmax - cmin +1|

3. test1/test3/Program.cs

```
        }
    }
}
```